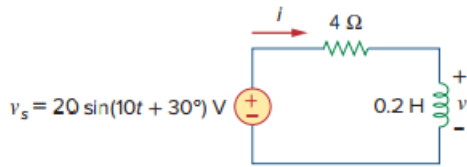


Problem

Refer to Fig. 9.17. Determine $v(t)$ and $i(t)$.

Figure 9.17:



Step-by-step solution

Step 1 of 5

Refer to figure 9.17 given in the textbook.

Consider the voltage source.

$$\begin{aligned} v_s &= 20 \sin(10t + 30^\circ) \\ &= 20 \cos(10t + 30^\circ - 90^\circ) \\ &= 20 \cos(10t - 60^\circ) \text{ V} \end{aligned}$$

Convert the voltage source into phasor form.

$$\mathbf{V}_s = 20 \angle -60^\circ \text{ V}$$

Step 2 of 5

Calculate the impedance $\mathbf{Z}$.

$$\mathbf{Z} = 4 + j\omega(0.2)$$

Substitute 10 for ω .

$$\begin{aligned} \mathbf{Z} &= 4 + j \times 10 \times (0.2) \\ &= 4 + j2 \, \Omega \end{aligned}$$

Step 3 of 5

Calculate current i in phasor form.

$$\mathbf{I} = \frac{\mathbf{V}_s}{\mathbf{Z}}$$

Substitute $20\angle -60^\circ$ for $\mathbf{V}_s$, and $(4 + j2)$ for $\mathbf{Z}$.

$$\begin{aligned}\mathbf{I} &= \frac{20\angle -60^\circ}{4 + j2} \\ &= \frac{20\cos(-60^\circ) + j20\sin(-60^\circ)}{4 + j2} \\ &= \frac{20\left(\frac{1}{2}\right) - j20\left(\frac{\sqrt{3}}{2}\right)}{4 + j2} \\ &= \frac{10 - j10\sqrt{3}}{4 + j2}\end{aligned}$$

Simply further.

$$\begin{aligned}\mathbf{I} &= \frac{10 - j10\sqrt{3}}{4 + j2} \times \frac{4 - j2}{4 - j2} \\ &= \frac{(10 - j10\sqrt{3})(4 - j2)}{4^2 + 2^2} \\ &= \frac{5.359 - j89.282}{20} \\ &= 0.2685 - j4.4641 \\ &= 4.472\angle -86.565^\circ \text{ A}\end{aligned}$$

Step 4 of 5

Convert current $\mathbf{I}$ into time domain.

$$\begin{aligned}i(t) &= 4.472\cos(10t - 86.565^\circ) \\ &= 4.472\sin(10t - 86.565^\circ + 90^\circ) \\ &= 4.472\sin(10t + 3.435^\circ) \text{ A}\end{aligned}$$

Therefore, current $i(t)$ is $\boxed{4.472\sin(10t + 3.435^\circ) \text{ A}}$.

Step 5 of 5

Calculate the voltage across the inductor.

$$\mathbf{V} = \mathbf{IZ}$$

Substitute $0.2685 - j4.4641$ for $\mathbf{I}$, and $j2$ for $\mathbf{Z}$.

$$\begin{aligned}\mathbf{V} &= (0.2685 - j4.4641)(j2) \\ &= 8.9282 + j0.537 \\ &= 8.944\angle 3.442^\circ\end{aligned}$$

Convert current $\mathbf{V}$ into time domain.

$$\begin{aligned}v(t) &= 8.944\cos(10t + 3.442^\circ) \\ &= 8.944\sin(10t + 3.442^\circ + 90^\circ) \\ &= 8.944\sin(10t + 93.442^\circ) \text{ V}\end{aligned}$$

Therefore, voltage $v(t)$ is $\boxed{8.944\sin(10t + 93.442^\circ) \text{ V}}$.